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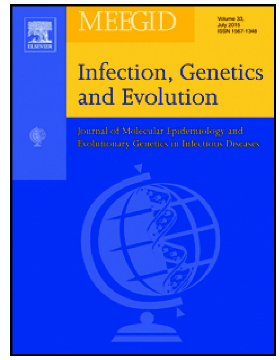

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Review Article

Evolving trends of New Delhi Metallo-betalactamse (NDM) variants: a threat to antimicrobial resistance

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Abstract

The rapid emergence of carbapenemase producing Gram-negative bacterial strains exhibit broad-spectrum β-lactam resistance, especially New Delhi metallo-β-lactamase (NDM-1). It is a major public health threat as it catalyses the hydrolysis of a vast variety of β-lactam antibiotics, including carbapenems, which is the last c [illegible] ide, in spite of constant efforts to control. Its clinical treatment remains challenging due to continuous evolution of new variants. A thorough structural study of all variants is required to develop new and effective inhibitors. This review focuses on the dissemination, position of substitution and carbapenemases activity of all the 28 NDM variants so far reported.

Keyword

Resistance, NDM, variants, inhibitor, β-lactam antibiotics

Introduction

A significant mechanism of resistance against broad-spectrum β-lactam is the production of carbapenemases in members of *Enterobacteriaceae* (Queenan and Bush, 2007). These carbapenemases can be characterised as acquired resistance as well as intrinsic resistance and are further subdivided into molecular classes A, B, C and D by Ambler classification (Queenan and Bush, 2007). These enzymes can also hydrolyze a specific type f different classes are evolving such as class A (GES, KPC), class B (VIM, IMP, NDM) and class D (OXA-23). Class C (CMY, PDC) are weak carbapenemases. However, its over-expression along with other mechanism of resistance can result in carbapenem resistance (Chia et al., 2009). Class B include Metallo-Beta-Lactamases (MBLs) which are efficient in hydrolysing nearly all clinically available β-lactam antibiotics. They have metallo-hydrolase/oxidoreductase superfamily distinct αβ/βα sandwich fold. MBLs narrow active-site groove contain divalent zinc ions (one or two) and is surrounded by flexible hairpin loop (Mojica et al., 2016). This flexible loop plays a significant role in hydrolysis by moving over the zinc ion and after catalysis it moves back to its original position in the millisecond time in NDM-1 (Aitha et al., 2016).

Further classification of MBLs into B1, B2 and B3 as sub-classes, is based on sequence identity and depends on zinc ion. The subclass B1 includes clinically most important enzymes. Among the designed inhibitors, limited number are successful because of the structural differences in the active site, catalytic mechanisms and nature of zinc ligands (Mojica et al., 2016). In Gram-negative bacteria β-lactamase inhibitors (BLIs) play a significant role in combating β-lactamase resistance. However, all the available β -lactamase inhibitors like clavulanate, avibactam, tazobactam and sulbactam are becoming ineffective due to continuously evolving variants and different variety of β-lactamases (Bush, 2015).

New Delhi metallo- β -lactamase (NDM), a type of metallo- β -lactamase (MBL), can hydrolyze most of the β-lactams (including carbapenems) except monobactams (Nordmann et al., 2011; Yong et al., 2009). The NDM-1 was first identified in *Klebsiella pneumoniae* which was isolated in New Delhi (India), from a Swedish patient hospitalized in 2008 (Yong et al., 2009). NDM-1 started emerging in various species of *Enterobacteriaceae* after 2008. NDM-producing strains are commonly resistant to almost all antimicrobial agents because of the co-occurrence of additional resistance mechanisms (Khan and Nordmann, 2012). Risk associated with NDM-1 strains are, unavailability of effective antibiotics, unrecognition of highly prevalent asymptomatic carriers, lack in the standardized routine phenotypic test for detection of metallo-beta-lactamase (MBL) and its presence on plasmids which has the ability to rearrange and transfer through horizontal transmission (Khan et al.,2017). *E. coli* which is producing NDM-1 invades the host at different sites y tract infections, soft tissue infections, device-associated infections, diarrhoea and peritonitis (Kaase et al., 2011). The virulence of these antibiotic resistant bacteria depends on specific clone having resistant marker associated with virulent factors, present on plasmids. NDM producing bacterial strain carrying plasmid, pNDM-BJ01 and pNDM-BJ02 comprises of both *bla* $_{\text{NDM-1}}$ and a type IV secretion system (T4SS) gene cluster showing high virulence. The T4SS belongs to the P-type T4SS group, that encodes a short and rigid pilus as one of the virulent factors (Hu et al., 2012). Mode of transmission of NDM-1 producing strains is through body fluids or cross-contamination of food. Community and hospital settings are also vulnerable for the spread of resistant strains of bacteria (Bogaerts et al., 2010). A total of 28 NDM variants have been reported so far.

Worldwide distribution

About 58.15% of NDM-1 and its variants are prevalent in Asian sub-continents especially in China, Bangladesh, Srilanka and India. Whereas, 16.8% abundance of NDM-1 and its variant are found in European countries. Almost 10.8% NDM-1 producers are reported to occur, each in USA and Africa, while Australia serves as the 1.6% of its reservoir. The percentage prevalence of the NDM-1 is out of the global presence of NDM-1 producers. Figure 1 shows abundance of NDM-1 and its variants in different part of the world. Highest prevelence of these NDM variants is detected in *E. coli* and *K. pneumoniae* species. (Khan et al., 2017).

Structure of NDM variants

NDM enzyme involved in hydrolysis of β-lactams, comprises of 17 -strands, 9 –helices and 3 turns (Fig.2). The NDM have 270 amino acids including two zinc ions at the active site, in which substitution has been reported at 17 different positions, resulting in 28 NDM variants. These variants normally contain 1 to 5 amino acid substitutions. The most common substitution is M154L, which is reported in 12 of the 28 NDM variants. NDM-1 and NDM-18 are similar; the only difference is 5 amino acids tandem repeat (QRFGD) at positions 44 to 48 of NDM-1(Fig 3) (Wu et al.,2019).

The mutation M154L present on the protein surface along with other mutations, is found to increase the stability of NDM variants through optimizing hydrophobic side-chain packing as shown by X-ray cryst… cing flexibility of the loop (G222D) or eliminating a charge from closely situated aspartate residues which might repel each other (D130N and D95N). The mutations that are involved in the stability of NDM hydrophobic core, specifically M154L, possible serve as global suppressors by introducing other mutations, that results in adding function at the cost of structure destabilisation. Global suppressing mutations have been reported in the serine β-lactamases evolution. (Huang et al.,1997; Patel et al.,2015; Brown et al., 2010)

It has been reported that structure of NDM variants is disturbed at the location of metal centre that results in change in the affinity of zinc leading to increase in resistance. All variants up to NDM-17(except NDM-1) has increased zinc affinity. However, change in structure of these variants also affect kinetic mechanism of enzyme, in addition to zinc affinity. It has been reported that some of the NDM variants functions as mono-zinc enzymes with same catalytic efficiency as di-zinc enzyme (NDM-1), because of uncoupling of the two zinc-binding sites by mutation resulting in altered kinetic mechanism. Amino acid substitutions are not observed in the active site of NDM variants (Fig. 2), but some variants have been found to have altered activities against β –lactams (Table 1). These variants do not show improved binding ability of di-zinc NDM or altered affinity of inhibitor when tested in zinc abundance (Cheng et al., 2018). They do not increase antibiotics resistance or induce large change in protein conformation in abundance of zinc. However, under zinc deprivation conditions, the NDM-4 (M154L), NDM-6 (A233V) and NDM-9 (E152K) are found to increase resistance against cefotaxime either by improvement in the metal affinity (M154L) or by improvement in the NDM enzymes stability (A233V and E152K). Whereas, NDM-3 (D95N) and NDM-14 (D130G) are also reported to enhance cefotaxime resistance under zinc starvation conditions, but mechanism of action is not yet known. No significant impact on NDM function is noticed on zinc deprivation in NDM-16 (R264H), NDM-11 (M154V), and NDM-2 (P28A). NDM-1 is evolved under stress to different variant which occurs under the condition of zinc deprivation (Bahr et al., 2018).

The variants containing mutations which have influence in the equilibrium of folded and unfolded states of the protein result in increased resistance. NDM variants have also been reported to have improved refolding efficiency as compared to NDM-1. The equilibrium of folded and unfolded states is reported to be measured by ca [...] here is increase in the values of Tm at the range of 10 °C (Cheng et al. 2018).

NDM-5, -17, -20, -21 and -24 variants which contain V88L substitution (Table 1), have been reported to show improved carbapenemase activity (Hornsey et al., 2011; Liu Z et al., 2017; Liu Z et al., 2018; Liu L et al., 2018; Liu Z et al., 2019). The MIC of ertapenem against NDM-5 and NDM-20 producing strains are reported to have 4 to 8 fold greater as compared to NDM-1 producing strains (Hornsey et al., 2011; Liu Z et al., 2018). NDM-17 and NDM-21 carbapenemase activity are found similar to NDM-5 (Liu Z et al., 2017; Liu L et al., 2018). This shows that such a substitution located in spite of non-active site, may have an important impact on enzyme activity. Other substitutions found to increase carbapenemase activity, are NDM-4 (M154L) and NDM-14 (D130G) (Nordmann et al., 2012; Zou D et al., 2015). However, M154L and D130G substitutions are found in NDM-8 but does not display any increased carbapenemase activity (Nordmann et al., 2012).

The study on NDM variants (NDM-1, -2, -4, -5, -6, -7 -8) do not define major differences in the catalytic efficiencies against carbapenem or penicillin substrates (Makena et al., 2015). A study reported that all the NDM variants (NDM-1, -3, -4, -5, -6, -7, -8, -9, -11, -12, -13, -14, -15, -16) have fully-occupied di-zinc active sites and show similar steady-state catalytic parameters. They all have k_{cat}/K_m values in the range of 10^6–10^7 M^{-1} s^{-1} indicating that they are efficient catalysts of β-lactam hydrolysis. (Cheng et al. 2018). Figure 4 shows relative resistance of NDM variants in zinc-depleted conditions. However, more study on catalytic efficiency and relative resistance of newly evolved NDM variants is need to be performed.

Artificially induced mutations in NDM5 and NDM7 have been generated in few non-active site residue to understand crucial role in enzyme function. NDM-7 flexible loop mutations S191A and E152A showed that these residues play a significant role in conferring β-lactam resistance and hydrolysis of the antibiotics. Moreover, the double mutant of NDM-7 (E152A+S191A) showed enhanced susceptibility as compared to the single-mutant (E152A/S191A), concluding the synergistic role of both residues in enzyme catalysis. In addition, S191 and E152 have also been reported to affect NDM-7 thermal stability directly or indirectly as revealed by thermal unfolding analysis of the protein secondary structure (Kumar et al. 2020). It is reported that E152A mutation at the probable omega-loop of NDM-5 has made the enzyme-harbouring cells susce[illegible]cial role in conferring beta-lactam hydrolysis and exerting resistance. The study shows that NDM-5 E152A mutation also affect the thermal stability by reducing the beta-sheet structure (Kumar et al. 2017). Moreover, our research group has also explored the role of several distant site residues of NDM-1 enzyme in its catalytic activity against carbapenems (Ali et al, 2019[a], 2019[b]), whereas, one mutation (Q123A) was found to be a rare to enhance the catalytic efficiency of NDM-1 against Carbapenems (Ali et al, 2018).

Conclusion

The emergence of NDM-producing bacterial strains has become a threat to public health. The worldwide spread of NDM-1 and its variants is caused by horizontal gene transfer through different types of plasmids and mobile genetic elements. The emerging trends of NDM-variants may lead a global crisis for health care worker and physician to treat deadly infections. Therefore, its structure analysis to understand its function is an utmost important task. This review described all possible structural changes occurred due to mutations in different NDM variants. After thorough understanding of the insight of structural and functional relationship, a potential inhibitor may be designed as a future therapeutic approaches which is the need of current situation.

Declaration of interest

The authors report no conflicts of interest.

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Fig 1: Worldwide prevalence of NDM-1 and all its variants.

Fig 2: Representation of the secondary structures in a primary sequence.

Fig 3: Comparison of all the NDM variant by superimposing on NDM-1 and showing substitution at different location.NDM-18 is showing structural difference because of additional 5 amino acid residues. All other mutations do not have much effect in structure and is showing almost same structure as NDM-1.

***colour scheme NDM-1 purple, NDM-2-sea green, NDM-3-yellow, NDM-4 aquamarine, NDM-5- olive green, NDM-6-magenta, NDM7-wine, NDM-8-red, NDM-9-peach, NDM-10-orange, NDM-11-brown, NDM-12-dark pink, NDM-13-dark red, NDM-14-navy blue, NDM-15-blue, NDM-16-green, NDM-17-ivory, NDM-18-lemon, NDM-19-tangerine, NDM-20-pink, NDM-21-crimson, NDM-22-light green, NDM-23-violet, NDM-24-grey, NDM-25-white, NDM-26-royal blue, NDM-27-light pink, NDM-28-lavender.

Fig 4: Representation of relative resistance of NDM variants in zinc-depleted conditions

Table 1: Genetic variations, source organisms, host, carbapenemase activity, zinc content and Tm range among the NDM-1 and its variants reported so far. (NA-data not available).

Ndm-1 variant	Gene substitution	Amino acid substitution	Location of amino acid substitution	Source organism	Host	Carbapenemase Activity	Zinc content	T_m (ºC)
NDM-2	C82G,A468G	P28A	region between lipidation box and β1-strands	*Acinetobacter baumannii*	Homo sapiens	Similar	NA	NA
NDM-3	G283A	D95N	α1-helix	*Escherichia coli*	Homo sapiens	Similar	1.9 ± 0.1	57
NDM-4	A460C	M154L	region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Increased	1.6 ± 0.1	58.7
NDM-5	G262T, A460C	V88L, M154L	β6-strands, region between	*Escherichia coli*	Homo sapiens	Increased	1.6 ± 0.1	62

			and Turn2					
NDM-6	C698T	A233V	α8-helix	*Escherichia coli*	Homo sapiens	Increased	1.8 ± 0.1	NA
NDM-7	G388A, A460C	D130N, M154L	α3-helix, region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Increased	1.6 ± 0.1	61
NDM-8	A389G, A460C	D130G, M154L	α3-helix, region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Similar	1.7 ± 0.1	61
NDM-9	G454A	E152K	Turn2	*Klebsiella pneumoniae*	Homo sapiens	Similar	1.6 ± 0.1	58.5
NDM-10	C94A, G107A, G2015A	R32S, G36D, G69S, A74T, G200R	β1-strands, Turn1, β5-strands region between β14-strands and β15-strands	*Klebsiella pneumoniae*	Homo sapiens	Increased	NA	NA
NDM-11	A460G	M154V	region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Similar	1.9 ± 0.1	56
NDM-12	A460C, G665A	M154L, G222D	region between α4-helix and Turn2, α7-helix	*Escherichia coli*	Homo sapiens	Similar	1.5 ± 0.1	59
NDM-13	G283A, A460C	D95N, M154L	α1-helix, region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Similar	1.4 ± 0.1	60
NDM-14	A389G	D130G	α3-helix	*Acinetobacter lwoffii*	Homo sapiens	Increased	2.0 ± 0.1	58.5
NDM-15	A460C, G698T	M154L, A233V	region between α4-helix and Turn2, α8-helix	*Escherichia coli*	Homo sapiens	NA	1.8 ± 0.1	61.5
NDM-16	G791A	R264H	α9-helix	*Klebsiella pneumoniae*	Homo sapiens	NA	1.9 ± 0.1	64

NDM 17	G508A	M154L, E170K	region between α4helix and Turn2, region between β11-strands and α5-helix	*coli*	n		0.1	60
NDM-18	-	QRFGD, amino acid positions 44 to 48 of NDM-1	β2-strands, region between β2-strands and β3-strands	*P. rettgeri*	Homo sapiens	NA	NA	NA
NDM-19	G388A, A460G, C698T	D130N, M154L, A233V	α3-helix, region between α4-helix and Turn2, α8-helix	*Escherichia coli*	NA	NA	NA	NA
NDM-20	G262T, A460CG809A	V88L, M154L, R270H	β6-strands, region between α4helix and Turn2	*Escherichia coli*	Swine	Increased	NA	NA
NDM-21	G205AG262T, A460C	G69S, V88L, M154L	Turn1, β6-strands, region between α4-helix and Turn2	*Escherichia coli*	Homo sapiens	Increased	NA	NA
NDM-22	A742C	M248L	region between β16-strands and β17-strands	*Enterobacter cloacae*	Homo sapiens	NA	NA	NA
NDM-23	A301C	I101L	*α1-helix*	*Klebsiella pneumoniae*	Homo sapiens	NA	NA	NA
NDM-24	G262T	V88L	β6-strands	*Providencia stuartii*	Homo sapiens	Increased	NA	NA
NDM-25	-	A55S	β3-strands	*Klebsiella pneumoniae*	NA	NA	NA	NA
NDM-26	G262T, A460C,	V88L,M154L,G222S	β6-strands, region	*Escherichia coli*	Swine	NA	NA	NA

			α4-helix and Turn2					
NDM-27	G283A, C698T	D95N,A233V	α1-helix, α8-helix	*Escherichia coli*	Homo sapiens	NA	NA	NA
NDM-28	-	A266V	*α9-helix*	*Klebsiella pneumoniae*	NA	NA	NA	NA

High

- NDM variants are evolving with improved carbapenemase activity, threatening to public health.
- Drugs and inhibitors are becoming ineffective due to continuous evolution of β-lactamases variants.
- Structural and functional study of all variants are important to design potential inhibitors.